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From colour target to ICC

InkProf - Quality profiling for RGB printers

Requirements and startup

Source version 1.2.1, released 2026-10-16; guide updated 2026-10-16. Tested with base MATLAB R2025b (no additional MATLAB toolboxes), Python 3.13 and ArgyllCMS on macOS. Python performs colour analysis and report generation; MATLAB provides the application and workflow. Install these runtimes separately. Windows and Linux are untested; use of a POSIX measurement bridge does not establish Linux compatibility. Run the commands below from the InkProf source folder.

How to use this guide

A step is a numbered row in the application’s left-hand workflow table. Follow the saved status and suggested next step. A completed calculation is evidence, not automatic approval of print quality.

```
setupInkProf(CheckPython=true);
startInkProfApp();
```


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One project, one source of settings

Use "New project" once, then "Open project" to resume. "Project details" contains the "Project and materials", "Printing settings", "Profiling" and "Target paper" tabs. Record printer, paper, ink/coating, application and driver settings, drying time, profile name, colour-data choices and paper/sled limits. Unknown values remain unknown. Shared project-controlled recipe fields are read-only in the recipe dialog. The separate "Compensate to D50 (also updates Project details)" FWA checkbox remains editable when spectral data are available; saving the recipe also updates the project’s FWA setting. Relevant changes invalidate dependent results while preserving earlier evidence.

Rename through "Project details" to rename the project folder too. External folder renaming is detected for confirmation. Move the entire project folder between computers; locally discovered Python and Argyll paths belong to the computer, not the portable project. "Verify project" checks recorded file hashes. "Close app" after saving/completing the active operation; reopen the same project after drying.

FWA and measurement conditions

FWA compensation models D50 effects from suitable M0 spectra. It does not convert an M0 instrument measurement into a native M1 measurement. Record the actual instrument conditions and compensation used. You can enable compensation later without changing code: retain native spectra, save a new B2 recipe, rebuild B3 and repeat checks. New physical verification is required for the changed profile.

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The 19 steps

Step	Action	What it creates or decides
1	Create/import RGB target	RGB definition and generation metadata
2	Save TIFF16	Matching TIFF16, TI2, layout and manifest; print separately
3	Measure/select revision	Instrument readings and saved JSON/TI3 revision
4	Review and accept	Explicit acceptance of the selected revision
5	B1: lock inputs	Immutable, checked profiling input snapshot
6	B2: recipe	Manual build recipe; project settings are reused
7	B3: build ICC / automatic iteration	ICC candidate, build arguments and job log
8	Fit, Grid and C1	Model fit, transformations and numerical checks
9	C2: save verification TIFF16	Profile-applied verification target
10	Measure verification	Saved verification revision

Build and check labels

B1 freezes validated input; B2 records the build recipe; B3 builds the ICC. C1 checks numerical transformations, inverse behaviour and CMM consistency. C2 produces a profile-applied print target; C3 compares its measured colours with desired colours and profile predictions. The Grid check samples an RGB lattice and checks round trips and ramps. Automatic iteration builds and evaluates candidates under its recorded selection rules; it does not approve a profile.

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The 19 steps - continued

Step	Action	What it creates or decides
11	C3: compare model and print	Desired vs measured and predicted vs measured errors
12	Review feedback	Recorded feedback/decision, not merely opening a window
13	Approve intended use	Explicit scope, notes and confirmation tied to current evidence
14	Save ICC and certificate	Delivery copies plus PDF/HTML measurement certificate
15	Refinement tools	Image, manual gamut/shell, errors or ICC-grid trials; saved refinement target
16	Measure refinement	Reviewable additional measurements
17	Build next iteration	Combined training set and a new ICC candidate
18	Compare previous/current ICC	Both ICCs evaluated at common RGB sample points; not dependent on their measurement grids
19	Save without print verification	Numerically checked later iteration and explicitly limited certificate

Ready: Required inputs are available. Select the step and "Run selected step" to start.
Complete: The step’s result or decision is saved and the app’s requirements for that step are met. This does not itself mean approved print quality.
Locked: The step cannot start yet. Read its description and first complete the prerequisite steps.
Stale: An earlier result needs updating because its inputs or settings have changed. Repeat the affected steps.
Use the dialog’s intended save or acceptance button. Simply opening and closing the dialog does not complete the step.
"Refresh" reloads saved status; it does not save edits still open in a dialog. Finish or cancel the active dialog before starting another operation.

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Targets, paper and measurement

Print the matching target

TIFF16 is saved, not sent to the printer. Use "Copy TIFF16 for printing" and read the accompanying PRINTING.txt. Print at 100% under the recorded conditions and allow drying. Base and refinement targets contain device RGB: disable further application/driver ICC conversion. C2 colours have already been converted once through the selected ICC; do not convert them again. Device-RGB reference patches in C2 retain their specified values. Keep matching TI2 and layout data in the project. Patch-row labels appear on both sides; compact layouts reserve a footer band.

Choose paper dimensions

"Target paper" suggests cut sheets A5 through A3+ or roll/custom sizes according to patch count and equipment limits. The 320 mm scan-width and 370 mm length defaults describe one sled, not universal instrument limits. Edit the limits in "Project details" and review the proposed layout.

Measure and review

Follow calibration and measurement prompts. Instrument identity is recorded when available. A remeasured patch remains a candidate until accepted and saved as a new revision. Select that revision and repeat dependent reviews/checks. A Lab value describes one measured colour; a change in Lab compares two readings of that patch. Optional ColorChecker reference targets can support comparison with familiar reference colours; their presence depends on the selected target and recorded evidence.

Instrument instructions

See docs/usage/instrument-startup.md and docs/usage/chart-measurement.md for calibration, reconnect and retry instructions.

Save and resume completed rows

Choose Save completed rows / pause and wait for confirmation before closing. A save requested during scanning waits for a row prompt and excludes the unfinished row; paired rows require both scans. Later choose Resume saved session and its folder, then Resume measurement. The dialog states page, row and direction. Confirm the SAME printed sheets; reprinted or replaced pages need a new session and fresh RGB checks. A checkpoint is not a completed workflow measurement. Real-instrument resume validation remains necessary.

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Refinement and shadows

Open step 15, "Refinement tools", to choose the source of additional patches.

Use image colours or C3 feedback to propose more patches. An image's embedded ICC is used; if absent, a warning precedes the sRGB fallback so you can choose another image. Estimated-error filtering defaults to ΔE00 > 5. These are predictions, not measured errors.

Unmeasured C2 colours can be included with the new target. C2 colour/gray/challenge patches train the next iteration; repeated and paper-white patches retain control roles. Once used for fitting they are not independent validation of that next ICC. Challenge patches deliberately probe difficult colours or transitions; they are not automatic proof of print accuracy.

For Matte paper, the configurable shadow strategy defaults to targen patch emphasis 2 (-V2), colprof grid emphasis 1.3 (-V 1.3) and up to 48 extra dark fitting patches per iteration. The first target has no current ICC: emphasis redistributes its chosen patch budget. Later refinement can use the current ICC to select the lower quarter of its black-to-white L* range and exclude RGB16 duplicates. Additional dark patches do not remove chosen image/C3 colours. Counts and source ICC identity are recorded. This is an InkProf strategy, not an official Argyll matte preset; improvement requires measurement. Existing targets remain unchanged.

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Shell refinement and continuation

Use Refinement tools to strengthen coverage near the gamut surface.

Review before printing

Choose the whole inner shell or selected areas, set geometric thickness in ΔE76, and review plotted candidates. Save the TIFF16 package and use its matching target for measurement. Saved packages are registered in step 15 without regenerating patch order.

Continue from measurements

Measure in step 16 and build in step 17. Step 17 runs Fit, Grid and C1 as part of the build. Compare in step 18; if the current checks are missing or stale, run step 8 before delivery or verification. Follow the app’s suggested next step. A build stopped by numerical alerts retains measurements and the previous active ICC. Open step 17 results and inspect the latest C1/Grid reports before retrying.

Choose the delivery path

With current numerical checks, choose either steps 9-11 for a new verification print and measurement, step 12 to review feedback, step 13 to approve intended use and step 14 to deliver; or step 18 for comparison and step 19 for explicitly numerical-only delivery. Earlier measurements are historical evidence, not verification of the new ICC.

Iteration: 16 -> 17 (includes Fit, Grid and C1) -> 18
If checks are missing/stale: step 8.
Print-verified delivery: 9 -> 10 -> 11 -> 12 -> 13 -> 14
Numerical-only delivery: 18 -> 19

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Interpret and deliver

Fit error against training data, numerical consistency and physical print verification answer different questions. Large desired-colour errors with small model-vs-print errors may reflect gamut limits; this alone does not prove the cause. Shared-sample profile comparisons show change, not which profile is more accurate. Compare in 2D starting at L*=50, or switch to 3D. View gamut shows a separate ICC-derived surface; HTML can rotate it, PDF shows a still view. BPC (black point compensation) is not applied to the diagnostic absolute-colour comparison.

Stable delivery names and HTML support

The project retains the checked ICC candidate. The proposed external filename combines the profile name, iter-N and the final eight characters of the iteration UUID; the save dialog lets you choose its destination. The same stem identifies the ICC and certificate. The delivery ICC description is updated, so its file hash differs from the checked candidate even though colour-tag payloads are preserved. Keep the complete delivery folder and its underlag support folder for portable HTML, linked reports and assets.

Certificate and separate measurement appendix

The main certificate contains scope, project/printing conditions, ICC identity, summary, available ColorChecker reference/measured pairs, decision and signature space. Historical ColorChecker results are labelled as such. Complete accepted measurements and technical details are in the separate measurement-appendix.pdf. Appendix A and Appendix B inside the certificate explain methods/standards and legal terms; they are distinct from that separate PDF. The measurement appendix is linked by certificate ID and the checked candidate’s SHA-256. The certificate separately records the named delivery ICC and its SHA-256.

Interpret standards carefully

Selected ISO 12647-7:2016 comparison limits are referenced through MediaStandard Print 2018, published by Bundesverband Druck und Medien (bvdm). This custom workflow is not ISO certification. Numerical-only delivery explicitly records the absence of separate print verification.

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Code, licensing and runtime checks

InkProf code and authored documentation: GNU GPL v3 or later, without warranty; see LICENSE and THIRD_PARTY_NOTICES.md. Dependencies and schema/fonts have their own terms. This is a source-only release: no MATLAB, Python or Argyll runtime is included. See docs/releases/v1.2.1.md and licenses/review-v1.0.0.md for the checked release boundary and remaining validation limits. The v1.0.0 licence review is a historical review, not a new legal clearance for every v1.2.1 change.

Version checks and reporting problems

InkProf checks required tool versions and Python imports during setup. Incompatible versions or unavailable imports can prevent operations. Use the setup diagnostics and saved job logs when reporting a problem through the public repository’s Issues page. This guide does not promise a support service.

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